

Path shadowing correction

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When shadowing corrections are applied (only to the CSAT3s) to data that passed QC , this is always done *before* any other processing: in these cases, other procedures such as calculation of planar fit rotation matrices and the rotations themselves, calculation of means and statistics, etc., are performed on the corrected data.

From the (u, v, w) in an orthogonal system measured by the CSAT3, we recover the path components (Horst and Oncley, 2006, Eq. (10)) from

$$u_a = -u \cos(\phi) + w \sin(\phi), \quad (1)$$

$$u_b = \frac{u + \sqrt{3}v}{2} \cos(\phi) + w \sin(\phi), \quad (2)$$

$$u_c = \frac{u - \sqrt{3}v}{2} \cos(\phi) + w \sin(\phi), \quad (3)$$

where $\phi = \pi/3$ is the angle between the CSAT3 paths and the horizontal. Next, we calculate the instantaneous absolute value of the (uncorrected) wind vector,

$$S = \sqrt{u^2 + v^2 + w^2}, \quad (4)$$

and then, for each path component, the angle between the wind vector and that path:

$$\theta_\alpha = \arccos(u_\alpha/S), \quad (5)$$

where $\alpha = a, b, c$. Then the path components are corrected by (Horst et al., 2015)

$$u_\alpha := \frac{u_\alpha}{0.84 + 0.16 \sin(\theta_\alpha)}. \quad (6)$$

where $:=$ stands for the usual assignment in computer languages. The orthogonal components are then recovered by an inverse rotation (Horst and Oncley, 2006, Eq. (9)):

$$u = \frac{-2u_a + u_b + u_c}{3 \cos(\phi)}, \quad (7)$$

$$v = \frac{u_b - u_c}{\sqrt{3} \cos(\phi)}, \quad (8)$$

$$w = \frac{u_a + u_b + u_c}{3 \sin(\phi)}. \quad (9)$$

Note that the newly recovered (u, v, w) is different from the raw measurement; therefore, one needs to iterate through (4)–(9) until convergence is achieved in all components. In this work, the convergence criterion is that the maximum difference over the three components is less than 0.001 m s^{-1} , which is well below the stated accuracy (0.01 m s^{-1}) of the CSAT3.

Campbell’s CSAT3B manual (Campbell Scientific, Inc., 2023, section 7.1.3) mentions that the option to apply the shadowing correction uses 3 iterations. Zhou et al. (2018) give a very detailed description of the shadowing correction algorithm, and impose, as done here, a convergence criterion of 1 mm s^{-1} on each velocity component. They report a maximum of 3 iterations until convergence. In our case we found that a maximum of 6 iterations were needed.

References

Campbell Scientific, Inc. *CSAT3B 3-D Sonic Anemometer with Integrated Electronics*, 04/2023 edition, 2023.

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Horst, T. W., Semmer, S. R., and Maclean, G. Correction of a non-orthogonal, three-component sonic anemometer for flow distortion by transducer shadowing. *Boundary-Layer Meteorology*, 155(3):371–395, Feb 2015. ISSN 1573-1472. doi: 10.1007/s10546-015-0010-3. URL <http://dx.doi.org/10.1007/s10546-015-0010-3>. corrections for sensor flow distortion.

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